

Group Project - Group 11

Members

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Introduction

"The European airport community is a key enabler of economic and social growth, resilience, security, and Europe's global competitiveness" states the European Commission's Directorate-General for Mobility and Transport. This suggests that disruptions within this European airport community can affect economic and social growth, resilience, security and Europe's global competitiveness. One of such disruptions is flight delays. Flight delays are a remaining issue that can have substantial negative consequences, such as an increase in operational costs for airports, enhanced environmental impact (Pastorini and Zanin, 2022), passenger dissatisfaction, and a negative impact on the perceived reliability of European airports. To mitigate these effects, a thorough analysis of the causes and prediction of flight delays is essential. Such analyses can yield valuable insights that support the optimisation of airport operations and the enhancement of their overall resilience.

Previous studies have already analysed the causes and effects of flight delays. For instance, Ashford et al. (2013) and Cohen and Paul (2003) showed how delays in operations at airports can lead to delays of dependent operations. This, in turn, may result in delayed flight departures (Cohen and Paul, 2003). Also, studies have examined external factors like weather (Rodríguez-Sanz et al., 2022) and air traffic management (Makhtoumi, 2023). Quantitative studies have analysed delay patterns and prediction models (for instance, see: Güvercin et al., 2021; Shao et al., 2019).

However, an integrated overview of the causes and predictions of delays at airports across Europe for recent, post-pandemic years has not been found. As a result, there is a gap in understanding of how airport-related factors, temporal and geographic variations, influence flight delay patterns in across Europe's airports in recent years.

Research Objectives

This project aims to address this gap by offering a **quantitative analysis of flight delays at the twenty European airports with the most flights**, during the years **2023** and **2024**. The twenty busiest airports are selected based on the dataset "AirportTraffic" provided by Eurocontrol. The busiest airports have been investigated as these might

have a high impact on the overall air traffic, and delays at these airports might have a larger impact. The years 2023 and 2024 are chosen as these years are assumed not to be impacted by the COVID-19 pandemic. The focus is on airport-related delays; en-route delays are out of the scope of this research. This decision is made as many delays are caused by inefficiencies at airports.

The research aims to answer the following main research question:

What are the main airport-related factors driving flight delays across the twenty busiest European airports in 2023–2024, and how accurately can these factors be used to predict the likelihood of potential flight delays through quantitative modelling and data visualisation?

To answer this research question, sub-questions are formulated:

1. What type of delays are occurring most at the airports?
2. Which airports experience the highest, the lowest delay, and most consistent delay rates?
3. How do delays relate to traffic volume?
4. How do delays distribute geographically across Europe, and which airports stand out as persistent "delay hotspots"?
5. What model can be used to predict whether an airport is prone to delays?

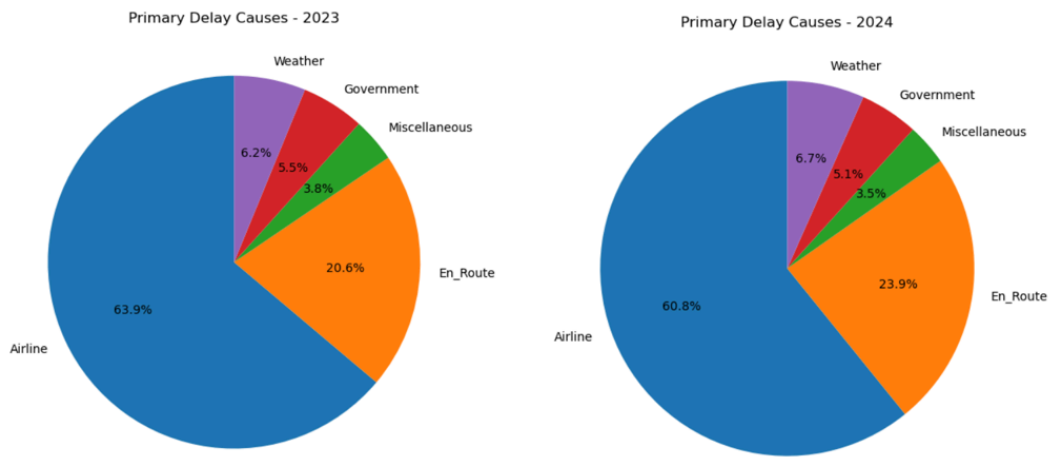
All datasets used in this research were obtained from Eurocontrol, a European organisation dedicated to supporting and coordinating aviation across Europe. As part of their commitment to transparency and research, Eurocontrol provides a wide range of publicly accessible data. Certain datasets, however, are restricted and available only upon request. One such dataset is the "IATA" dataset, which serves as a primary source for this study. The remaining datasets utilised are publicly available.

RQ1 - What type of delays are occurring most at the airports?

We identified the top 20 airports based on the total number of flights they handled in both 2023 and 2024, and saved them in a separate dataframe along with their respective ICAO codes. These codes will also be used in subsequent visualizations to provide airport-specific insights. Furthermore, this questions investigates the composition of flight delays of those twenty busiest European airports.

Primary delay causes

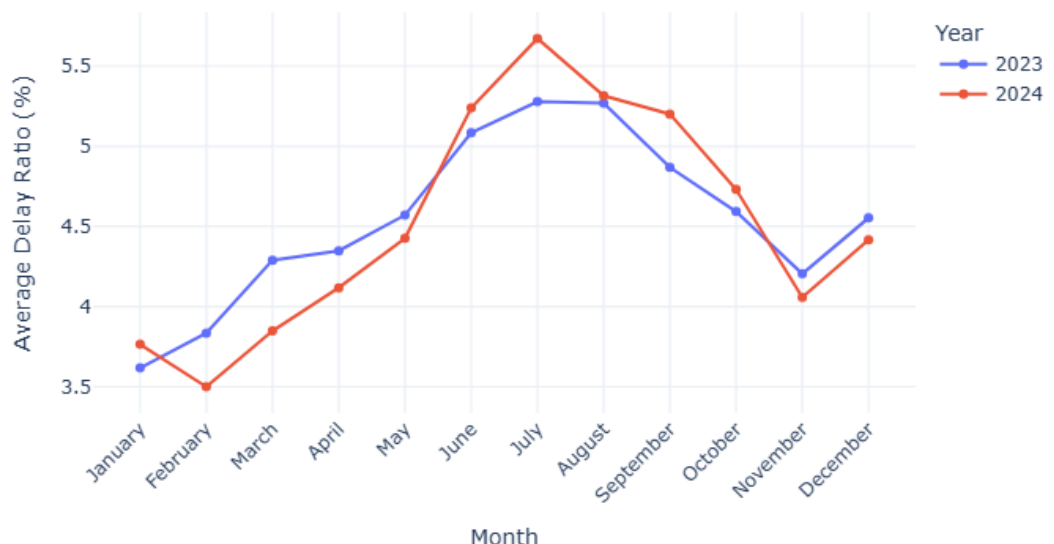
We start from an high-level where we provide insights into the major delay categories in general for all top 20 airports. The primary categories analysed include "weather" "government", "en_route" and "airline" where the following pie chart provides insights into the distribution of the delay causes for both 2023 and 2024.



IATA delay data analysis

In the aviation industry, the International Air Transport Association (IATA) has standardized a set of delay categories that enable consistent reporting and benchmarking across airports. Eurocontrol provides these delay reasons for all major European airports, allowing structured comparison of how different operational and external factors contribute to total delay. The following figure provides insights into the delay ratio per month for the airports comparing 2023 and 2024. It also contains a dropdown bar to select specific airport codes to understand airport specific delay ratios.

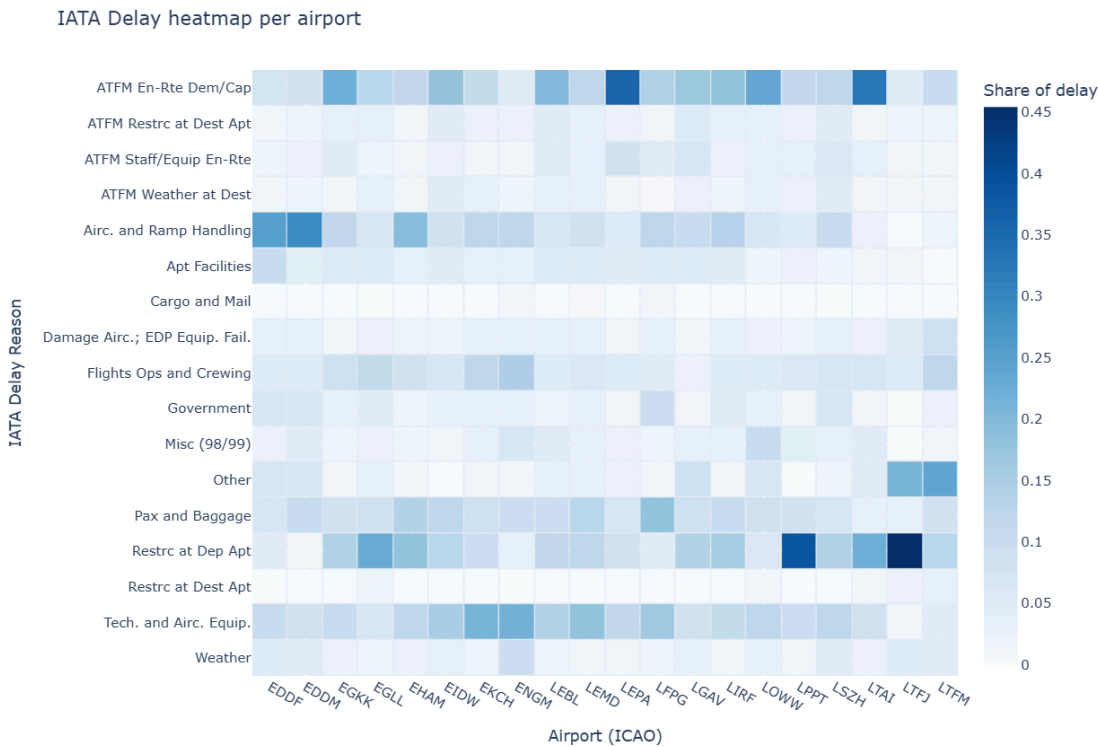
Average Delay Ratio by Month — All Airports



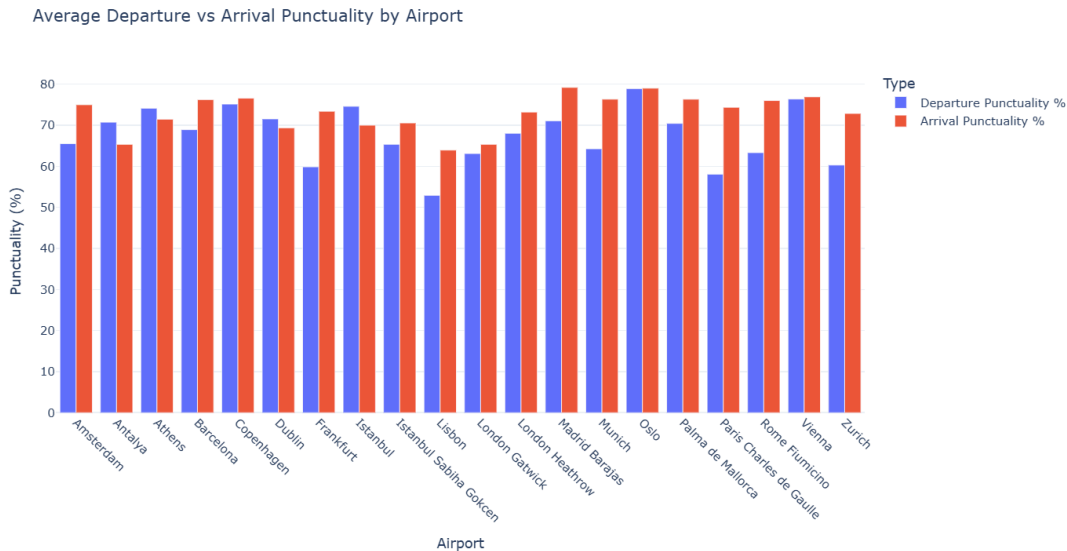
When analyzing the trendline, we can conclude that there is a seasonal delay peak happening in the summer. The average delay ratio climbs from 3.5% in winter to 5.5% in summer. When comparing 2023 with 2024, we can see some differences but overall we see a consistent trend.

The delay heatmap below presents the distribution of IATA delay reasons across the top 20 European airports, showing how each airport's total delay minutes are divided among

standardized categories. Darker shades represent a higher share of a particular delay reason at that airport.



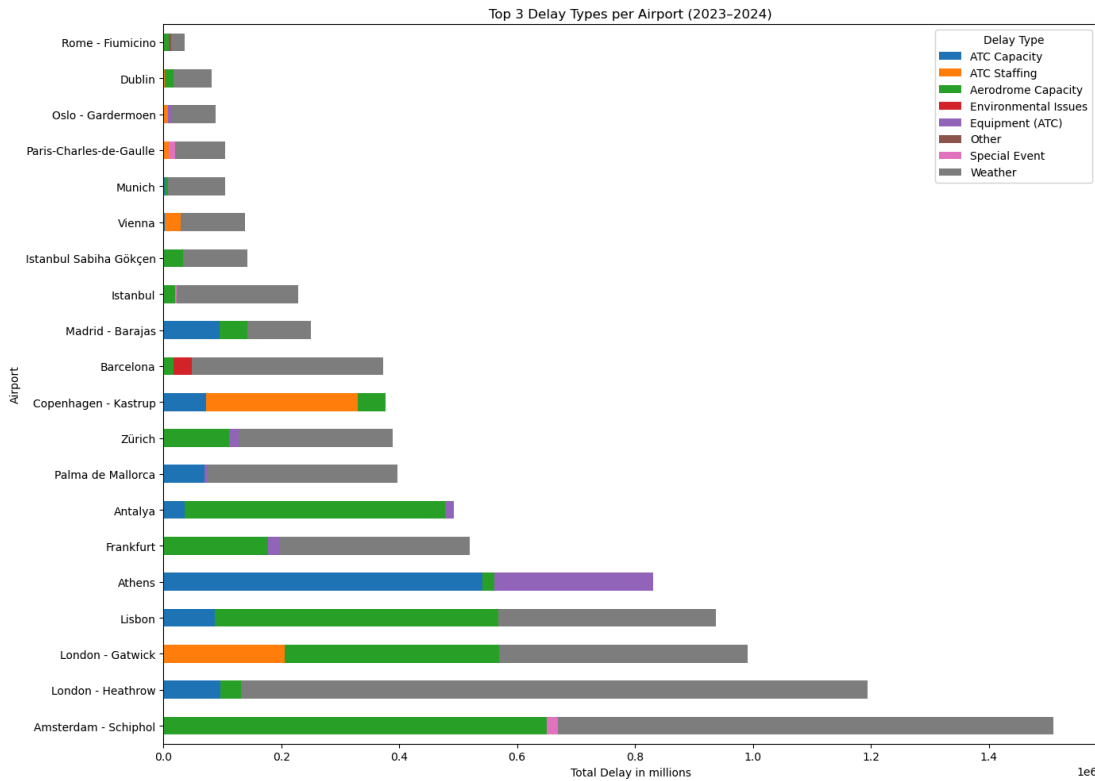
The airport arrival and departure punctuality are also analyzed for the airports to understand if it happens upon arrival or departure. The figure below compares average departure and arrival punctuality across airports. Each bar represents the proportion of flights departing or arriving on time, based on Eurocontrol's performance data.



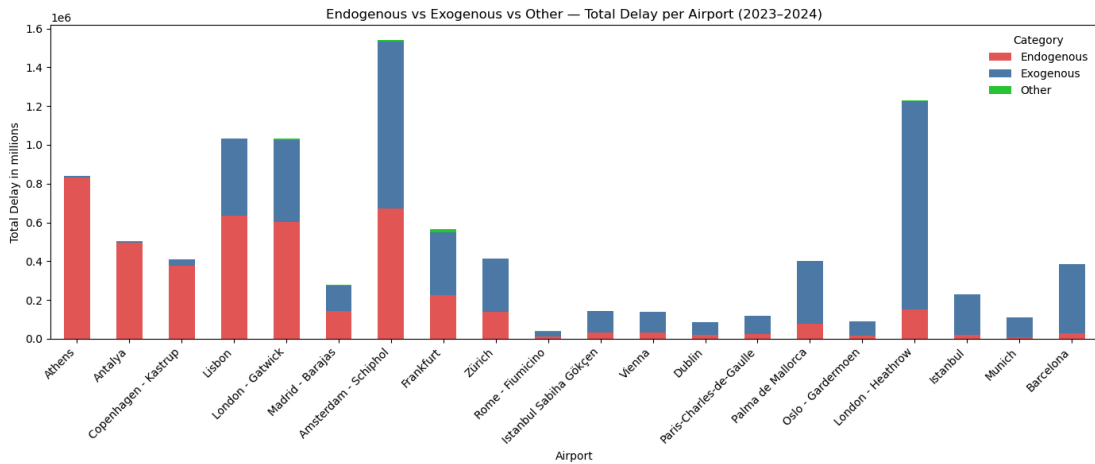
Overall, the chart shows that arrival punctuality is consistently higher than departure punctuality, suggesting that airlines often recover part of the delay during flight operations. Airports such as Munich, Oslo, and Zurich achieve the highest punctuality rates (around 75–80%), while Lisbon, Istanbul, and Palma de Mallorca exhibit lower departure performance, likely due to congestion and ground-handling constraints.

ATFM: Air traffic Flow Management Analysis

Diving deeper into the delays that only relate to air traffic, there are still some interesting trends and insights. The first major insight is related to weather. Where in IATA delays, weather is a small fraction of the delays. Weather is the biggest delay across all airports. But still varies a lot in the ATFM data set. Interesting is the amount of delays as well, major airports have very little delays related due to air traffic.



A comment about the wide difference between the largest and biggest amount of delays, a lot of airports like; Istanbul, Rome and Paris, have a lot of missing data or delays on zero. But nonetheless dividing the type of delays into two parts endogenous and exogenous shows another story as the relation between airports small or large, missing data or not, is visible. London Heathrow performs, certainly with its high traffic volume, exceptionally well at air traffic delays, as weather is the biggest cause. Even Schiphol is outperforming a lot of airports. Athens, Antalya and Copenhagen have a lot of work to do as almost all their delays are endogenous.

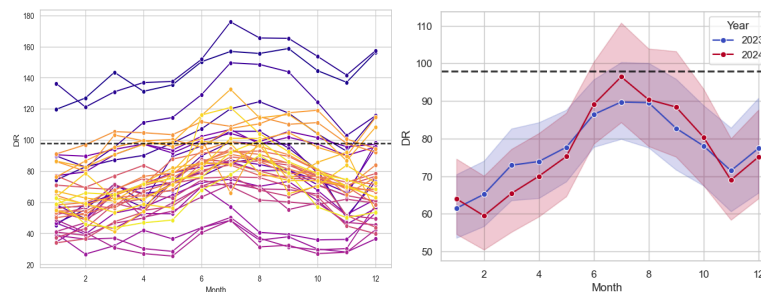


RQ2 - Which airports experience the highest, the lowest delay, and most consistent delay rates?

To answer the research question, we analyse the delay rate for each of the top 20 airports and, to show which airports face the most consistent delay rates, we use a Coefficient of Variation (CV).

Firstly, the delay ratio is given by the ratio of the number of flights affected by delay and the total number of flights in that period. This indicator was then replicated using the minutes of delay instead of the flights delayed.

The first delay ratio was used to compare the airport's performance and derive a reasonable threshold that would allow us to classify an airport as "Delay Prone" or not. From the charts below, we decided to use 98% since it's also a reference value for delay rates in the aviation industry.



The second variant of the delay ratio was used in combination with the CV to specifically answer the research question. The Coefficient of variation is calculated by dividing the standard deviation of this delay rate by its mean.

Monthly delay rate (min/flight) per airport (sorted by consistency value (cv))



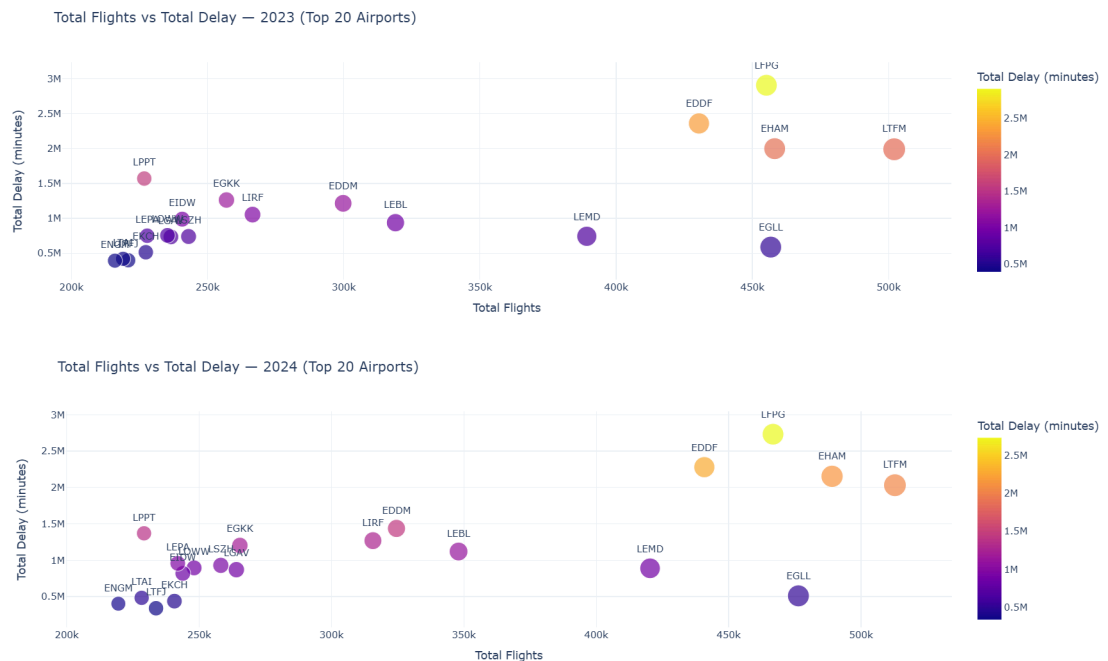
Based on the calculated and plotted monthly delay rates (based on the amount of delay minutes):

- **Airport with lowest monthly delay rate:** Oslo - Gardermoen experiences the lowest monthly delay rate in 2023 and 2024 out of the twenty busiest airports in Europe. This can be seen in the figure above: the mean delay rate is the lowest out of the examined airports (i.e. below 10 min/flight).
- **Airports with highest monthly delay rate:** Lisbon faces, on average, the highest delay rates.
- **Airport with most consistent delay rate:** Frankfurt has the most consistent delay rates as the coefficient value is the lowest (CV = 0.19) out of all airports. So, Frankfurt has the most stable delay rates from the twenty busiest airports in Europe.
- **Airport with most inconsistent delay rate:** On the contrary, the highest coefficient value (CV = 0.58) is indicated for Antalya. This means Antalya faces the most volatile delay rates. I.e.: the consistency of delay rates is the lowest at Antalya airport.

RQ3 - How do delays relate to traffic volume?

This research question explores the relationship between airport traffic volume and the total amount of delay experienced across the twenty busiest European airports in 2024. The analysis aims to identify whether higher flight activity leads to greater cumulative delays, and to what extent specific airports deviate from this overall trend.

Using Eurocontrol data for 2023 and 2024, total flight movements were plotted against total delay minutes for each airport. Figures below visualizes this relationship, where each point represents an airport; its position on the x-axis reflects total flights, the y-axis shows total delay (in minutes), and the colour gradient indicates the intensity of delay. The bubble size is proportional to total traffic volume, highlighting airports that dominate European air transport.



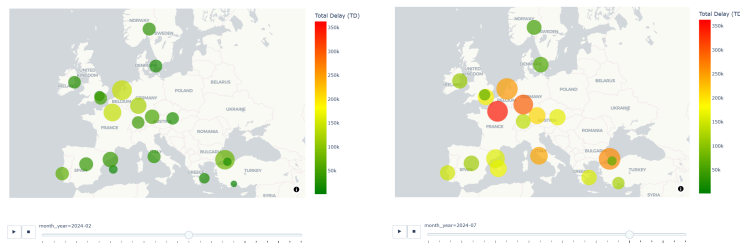
The scatter plot reveals a positive relationship between flight volume and total delay, confirming that airports handling more flights generally accumulate higher overall delays. You see also a similar pattern for both 2023 and 2024 which means there is constancy in the data sets. However, this association is not perfectly linear: several airports stand out as over- or under-performers relative to their traffic levels.

- **Congested high traffic airports:** Major hubs such as Istanbul (LTFM), Amsterdam (EHAM), Paris Charles de Gaulle (LFPG) and Frankfurt (EDDF) handle between 450 000 and 500 000 flights annually, resulting in total delay minutes exceeding 2 million. They show particularly high cumulative delays, reflecting congestion and operational complexity at large intercontinental hubs.
- **Efficient high-volume performers:** Madrid (LEMD) and London Heathrow (EGLL) manage very high traffic volumes but maintain comparatively lower total delays than peers of similar size
- **Medium underperformers:** Munich (EDDM), Rome (LIRF) and Lisbon (LPPT) have higher delay total (around 1 to 1.5 million minutes) despite its lower to moderate traffic volumes. This indicates potential capacity limitations or operational inefficiencies.

- **Smaller but resilient airports:** Airports such as Manchester (EGCC), Vienna (LOWW), and Brussels (EBBR) operate fewer than 250 000 flights yet maintain total delays below 0.6 million minutes, reflecting operational efficiency at moderate capacity.

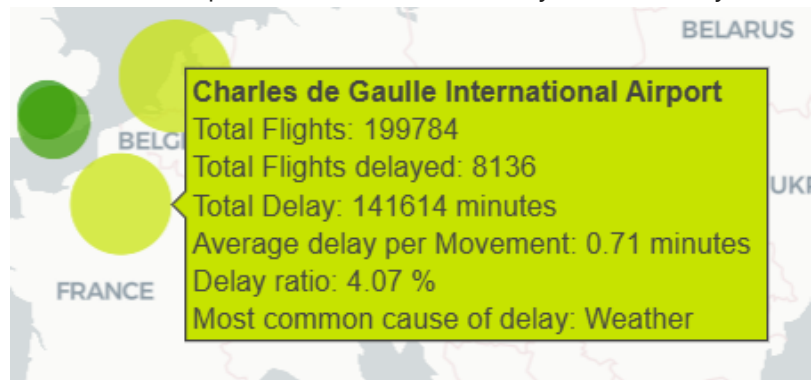
RQ4 - How do delays distribute geographically across Europe, and which airports stand out as persistent “delay hotspots”?

In the previous research questions, extensive data analysis was conducted. To present these insights in an even more visual and comprehensible manner, the 20 busiest airports in Europe were plotted on a map. This visualisation allows for a direct comparison of airport performance. The size of each point represents the total number of flights, while the color indicates the total delay in minutes. When hovering over an airport, detailed information about the delay at that specific airport is displayed. Additionally, a time slider has been incorporated, enabling users to easily observe how airport traffic and corresponding delays change over time.



The two screenshots above show the map visualizations for February 2024 and July 2024.

These illustrate how airport performance changes between winter and the summer holiday season, when air travel activity increases significantly. In July, more airports appear



in red, indicating longer delays. The dots are also larger, reflecting a higher number of flights, particularly in southern Europe, where many popular holiday destinations are located. This correlation is logical: as the number of arriving and departing flights increases, total delays also tend to rise. When hovering over an airport, key information about its performance is displayed.

RQ5 - What model can be used to predict whether an airport is prone to delays?

We used two different prediction models to answer this question: **logistic regression** and **random forest regression**.

Data preparation

We selected predictors based on their availability in the preceding month. For example, we excluded the total number of delayed flights for the target month because that data would not be gathered until after the month began. For both models, we will use: *month*, *total number of flights*, *average delay per movement*, *consistency value*.

The target variable is called "Delay_Prone" and is 1 if the airport is facing a delay rate above 98% during that month. The classification threshold was determined and commented on Research question 2.

After selecting the predictors and defining the target variable, we need to make sure they aren't correlated with each other.

Since the predictors aren't strongly correlated we can proceed with the logistic regression. The training the model and fitting the data the confusion matrix is:

	Month_Lobt	Total_Flights_Period	adm	CR	Delay_Prone
Month_Lobt	1.000.000	0.116096	0.204364	-0.004569	0.111887
Total_Flights_Period	0.116096	1.000.000	0.364938	-0.480967	0.191841
adm	0.204364	0.364938	1.000.000	-0.122351	0.478052
CR	-0.004569	-0.480967	-0.122351	1.000.000	-0.199941
Delay_Prone	0.111887	0.191841	0.478052	-0.199941	1.000.000

Logistic Regression

The logistic regression model performances are:

Predicted Label	Negative (0)	Positive (1)
Actual Negative (0)	68 (True Negatives)	13 (False Positives)
Actual Positive (1)	3 (False Negatives)	12 (True Positives)

Class	Precision	Recall	F1-Score	Support
0	0.96	0.84	0.89	81
1	0.48	0.80	0.60	15

To avoid bias related to the random state choice, we evaluated the model with 100 of them: from 0 to 99. The accuracy of the model was: 0.78, meaning that it is not sensitive to random state choice.

Finally, the logit formula that describes the model is:

$$\text{logit}(p) = -0.830 + 0.116 * \text{Month_Lobt} + -0.232 * \text{Total_Flights_Period} + 1.724 * \text{adm} + -0.872 * \text{CR}$$

The strength of the coefficients is:

Feature	Coefficient	Absolute Strength
adm	1.723.584	1.723.584
CR	-0,871926	0.871926
Total_Flights_Period	-0,232188	0.232188
Month_Lobt	0.115786	0.115786

Random Forest

The second model proposed is the random forest: we used the same parameters as in the logistic regression. In this case however, we split the dataset by airports: 16 will be used for training and 4 for testing. This is done because, with such few observations per airport, there is a high risk that the random forest "memorizes" patterns for each airport, rather than predicting them.

Given the same random state as before, the confusion matrix for the random forest model is:

Predicted Label	Negative (0)	Positive (1)
Actual Negative (0)	72 (True Negatives)	4 (False Positives)
Actual Positive (1)	14 (False Negatives)	5 (True Positives)

Class	Precision	Recall	F1-Score	Support
0	0.84	0.95	0.89	76
1	0.56	0.26	0.36	19

To avoid bias related to the random state choice, we evaluated the model with 100 of them: from 0 to 99. The accuracy of the model was: 0.81, meaning that it is not sensitive to random state choice.

Finally, the importance of the coefficient is:

Feature	Coefficient
adm	0.459419
CR	0.268151
Total_Flights_Period	0.224858
Month_Lobt	0.047573

When comparing the 2 models there isn't a big difference in accuracy: both can predict correctly the state of the airport. However, due to the small dataset and the behaviour of those mathematical models, logistic regression is probably the preferred choice since it's not able to memorize airport patterns due to its linear nature. Additionally, the logistic regression model can distinguish better the actually delay prone airports, having a Recall of 80% compared to the random forest 26%.

Conclusions

This study set out to identify the main airport-related factors driving flight delays across the twenty busiest European airports in 2023–2024 and to evaluate how accurately these factors can predict delay likelihood through quantitative analysis and modelling. We can address the main research question: “What are the main airport-related factors driving flight delays across the twenty busiest European airports in 2023–2024, and how accurately can these factors be used to predict the likelihood of potential flight delays through quantitative modelling and data visualisation?”

The analyses conducted throughout this study provide valuable insights into the causes and patterns of flight delays at Europe’s busiest airports. The airport flight delays are driven primarily by airline, airport, and en-route factors, while weather and government-related causes contribute to a smaller proportion of total delay minutes. However, within air traffic flow management data, weather emerges as the dominant cause, highlighting its growing impact under certain conditions. When looking into delay occurrences, a clear seasonal pattern is evident, with delay ratios peaking during the summer months when traffic volumes are highest. Delay ratios increase from approximately 3.5% in winter to 5.5% in summer.

In terms of punctuality performance, arrival times are more reliable than departure times, suggesting that airlines often manage to recover delays during flight operations. The delay intensity and consistency analysis revealed that Oslo - Gardermoen had the best performance, consistently maintaining the lowest average monthly delay rate (below 10 min/flight), while Lisbon faced the highest average rates. In terms of predictability, Frankfurt demonstrated the most stable performance, whereas Antalya showed the most volatility, indicating the greatest inconsistency in its monthly delay rates.

A positive relationship was observed between flight volume and total delay minutes, confirming that high traffic typically lead to greater cumulative delays. However, this relationship was not perfectly linear. Major hubs such as Istanbul, Amsterdam, Paris Charles de Gaulle, and Frankfurt face extensive congestion, while London Heathrow and Madrid-Barajas manage similar volumes more efficiently.

Geospatial visualizations highlighted clear seasonal and regional variations. During the summer months, airports across southern Europe showed significant increases in both flight activity and total delay time and average delay per movement. This pattern aligns with the tourism peak happening in the summer holiday season.

Two predictive models — logistic regression and random forest regression — were developed to forecast whether an airport would be “delay-prone.” Both models achieved strong predictive performance, with accuracies of 78% and 81%, respectively. Logistic regression was identified as the more interpretable and reliable model, particularly given the dataset size and its linear relationships.

Discussion

Overall, the findings for the research question are satisfactory and provide meaningful insights into the topic. However, analysing delays in aviation involves a wide range of

interrelated factors, making it a complex issue that would require years of comprehensive study. For the purpose of this research, the scope was therefore narrowed to a selection of datasets made publicly available by Eurocontrol. Nonetheless, public data often contain significant gaps. For instance, the ATFM_delay dataset reports daily delays per airport, but many entries are missing, which can lead to incomplete or biased results. This limitation became apparent during the analysis, prompting a request for access to an additional, non-public dataset, which proved to be much more complete.

However, a considerable portion of the analysis code had already been developed based on the earlier, incomplete datasets. Adapting it to the new dataset proved challenging and, in some cases, not entirely feasible, as the new dataset, while more complete, contained less detailed information overall compared to the original ones.

Another challenge encountered during the data collection process was the inconsistency between datasets. In some cases, the total number of daily flights differed across datasets, even though all were provided by Eurocontrol. Consequently, combining multiple datasets may not always produce a fully accurate or realistic representation, as discrepancies between reported figures can affect the validity of the results.

For future research on aviation delays, it would be valuable to examine a broader range of variables, including weather patterns, air traffic management efficiency, and airport infrastructure capacity. Combining these with more complete, and more detailed data could provide a more comprehensive understanding of the underlying causes.

Contribution Statement

Everyone:

- Background research and data analysis on data sources.
- Developing the storyline for the research question under their primary responsibility. Please note that the research questions were mainly led by the person indicated, but team members supported each other on work.

Hylke Bleeker:

- Data processing, analysis and visualisation for research question 1.

Mike Geerts:

- Introduction, conclusion writing after collaborative formulation of proposal with research questions.
- Data processing, analysis and visualisation for research question 2.

Jan-Pieter Vermeer:

- Data sources gathering from Eurocontrol

- Data processing, analysis and visualization and writing for research question 1.1 and 3

Bas Van den Muijsenberg:

- Data processing, analysis and visualisation for research question 4.
- Set up GitHub repo for the project.

Alessandro Casati:

- Data processing, analysis and model evaluation for research question 5.
- Writing report

Data Sources

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